

INF2009F - 2023
Draw a State Machine Diagram for a Circle Class
at Khwela Womxn
Exercise 10 - (15 Marks)

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IMPORTANT NOTE:

- Copied or Plagiarised answers will get 0 and student will be referred to Elsje Scott or Ayanda Pekane for further actions
- Please upload your completed exercise on Vula by the 2nd May 2023 at 9am (9:00)

Draw a state machine diagram that will depict the object states that a Circle class can be in. These statuses are important for reporting, decision making and to easily identify those circles that are highly successful and whose progress has been exceptional.

Exercise Task --- State Machine Diagram

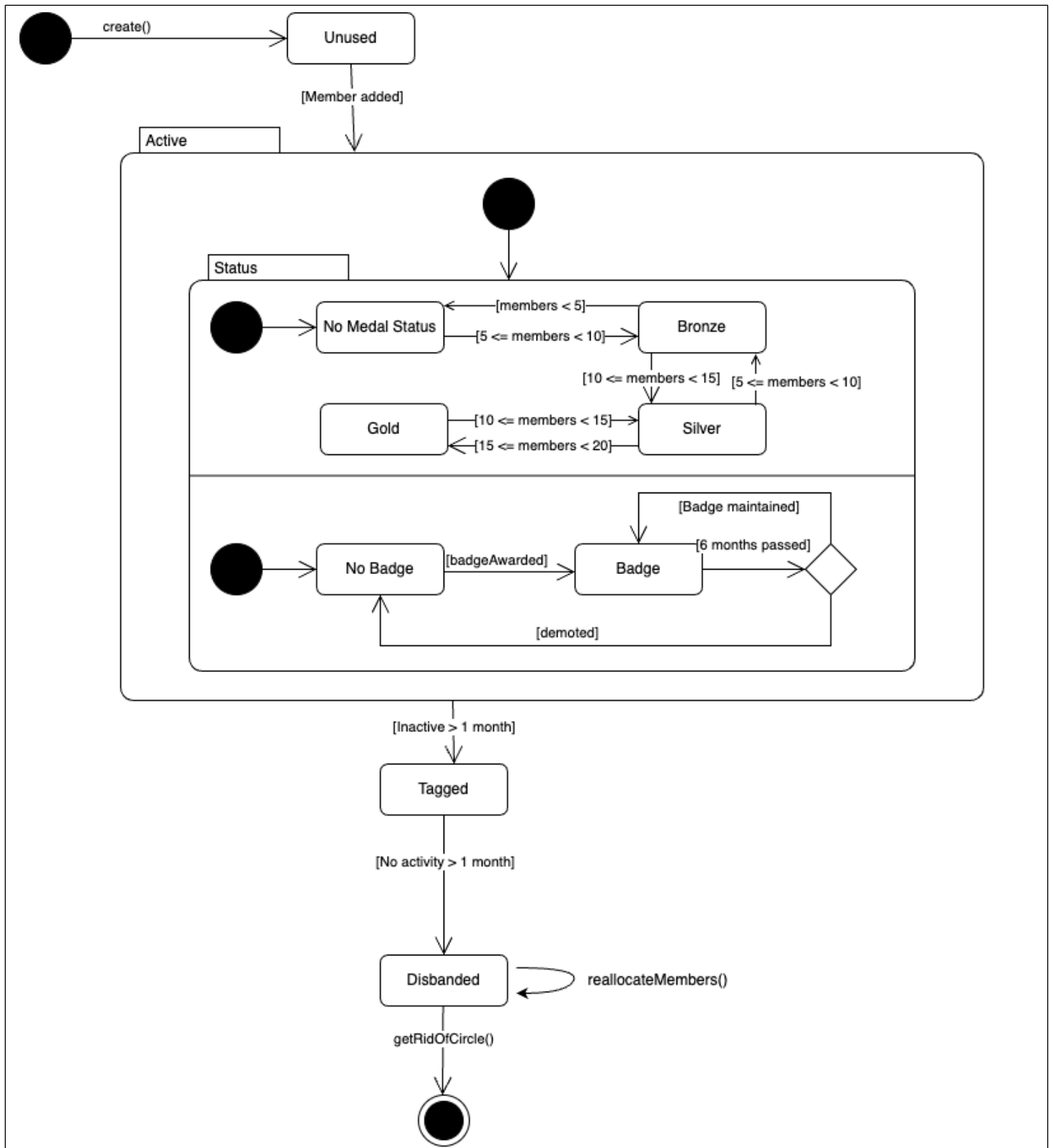
Draw a state machine diagram that will depict the object states that an object of a Circle class can be in. These statuses are important for reporting, decision making and to easily identify those circles that are highly successful and whose progress has been exceptional.

Additional information:

- When a request for a Circle is received the Khwela Womxn Team (KWT) will need to approve the request (Once the request is approved, they will allocate a moderator to manage and run the Circle).
- A Circle is active when it has Circle members on board.
- The KWT decided that circles will be assigned a bronze, silver or gold status depending on the number of members in the Circle (bronze $\equiv$ 5 or more members, but less than 10; silver $\equiv$ 10 or more, but less than 15; gold $\equiv$ 15 or more but less than 20).
- The KWT also wants to incentivise the circles by giving them a badge for having an impact or for attendance record that are exceptional. These badges will be assessed every 6 months to decide whether the Circle will maintain their status or go to a nobadge state.
- However, if the circles have been inactive for more than a month they will be tagged for investigation and facilitation.
- Finally, circles could be disbanded if no activity has been taken place for more than 1 month.

Provide a brief description of the state machine diagram.

Marks: Diagram - 14 Marks; Narrative - 1 Mark



A circle is in an Unused state when first created, and when a member is added, it is then in an Active state. An active circle has a status associated with it. The status involves both a medal status and badge status. Initially, a circle starts with no medal status, and if it gets from 5 to 9 members, it gets bronze status, silver for 10 to 14, and gold for 15 to 19 and a circle can go backwards and forwards through the medal rankings as they gain and lose members. Concurrently, a circle starts with no badge and if they get awarded a badge, their status changes to having a badge. After 6 months, this status is reviewed and either demoted to no badge if deemed so, or they maintain their badge. Then if an active circle is inactive for more than a month, they get tagged. Any tagged circle which is deemed to have no activity for a month is disbanded, its members reallocated and the circle removed.